ChandaGyan: Context-Aware Verse Correction and Meter Identification for Sanskrit Verses Using Fine-Tuned Masked Language Model



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By

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1. Introduction

1.1 Background

Metrical analysis (*Chanda Śāstra*) forms the computational and structural backbone of classical Sanskrit literature. Traditional computational tools, such as the *Chandojñānam* engine (Terdalkar, H., & Bhattacharya, A., 2023), rely strictly on deterministic, rule-based algorithms to parse character weight sequences (*Laghu* and *Guru*) and map them to fixed metrical patterns.

While deterministic parsers achieve exact accuracy on faultless inputs, they are inherently brittle. Minor typographic errors, OCR artifacts, dialectal variations, or missing syllables often lead to complete parsing failures. Recent advances in Natural Language Processing (NLP)—specifically Masked Language Models (MLM)—offer a probabilistic mechanism to model context, recover corrupted tokens, and infer structural intent. This study proposes re-implementing the core *Chandojñānam* engine and augmenting it with a Machine Learning (ML) verse-correction architecture to handle noisy input and suggest nearest-matching *Chandas*.

1.2 Objectives

1.2.1 General Objective

To enhance the *Chandojñānam* metrical analysis framework by integrating a machine learning-based verse corrector and nearest-match suggestion system.

1.2.2 Specific Objectives

- To implement and modernize the core rule-based engine of the *Chandojñānam* paper for baseline comparison.
- To design and fine-tune a Masked Language Model (MLM) to perform context-aware verse correction and token prediction on classical verse inputs.
- To develop a nearest-match retrieval module that maps corrupted or incomplete verses to their most probable *Chanda* pattern.
- To evaluate and compare the robust error-handling capability of the proposed hybrid ML approach against the traditional rule-based parser.

1.3 Rationale

Rule-based systems enforce rigid structural matching. When presented with real-world digital or digitized text containing minor OCR errors or spelling variations, traditional parsers fail entirely. Incorporating a Masked Language Model enables context-aware token recovery before or during metrical classification. By prioritizing deep ML architectural enhancements over broad language expansion (e.g., scoping out broad unannotated Nepali poetry to focus intensely on ML model efficacy), this project delivers a robust, error-tolerant solution for computational prosody.

1.4 Limitations

- The study focuses specifically on classical Sanskrit verse structures (*Chanda* rules) and fine-tuned linguistic models; general unannotated modern Nepali poetry corpus processing is excluded from the current scope.
- Model evaluation is constrained by the availability and quality of digitized, metrically annotated Sanskrit verse datasets.

2. Literature Review

The computational analysis of classical Sanskrit prosody (*Chanda Śāstra*) historically relies on deterministic, rule-based algorithmic models. Early digital implementations, such as the foundational *Chandojñānam* engine, operationalize traditional metrical principles by mapping text sequences into binary representations of light (*Laghu*) and heavy (*Guru*) syllables. These syllables are grouped into traditional *Gaṇas* to identify fixed metrical meters. While deterministic parsers achieve near-perfect accuracy when processing error-free, standardized inputs, their architecture is fundamentally brittle. Because rule-based engines depend on strict string matching against predefined prosodic templates, minor orthographic inconsistencies, missing characters, or Optical Character Recognition (OCR) artifacts lead to immediate parsing failures (Mishra et al., 2019; Hellwig, 2020). Consequently, traditional engines cannot gracefully degrade or infer metrical intent when presented with imperfect text.

To address the limitations of rigid parsing, recent research in Natural Language Processing (NLP) for low-resource and ancient languages has turned toward probabilistic and deep learning architectures. Transformer-based models, particularly Masked Language Models (MLMs) like BERT (Devlin et al., 2019), have demonstrated remarkable capabilities in understanding context and modeling sequence likelihoods. Specialized domain models, such as SanskritBERT (Sandhan et al., 2021) and broader multilingual representations like IndicBERT (Kakwani et al., 2020), leverage large-scale pre-training to capture fine-grained contextual and morphological dependencies in Sanskrit and related Indo-Aryan languages. By masking specific tokens during training, MLMs learn to predict corrupted or missing characters based on surrounding contextual constraints. Applying MLM architectures to metrical texts offers a robust mechanism for sequence reconstruction and token-level error recovery prior to prosodic classification.

Building upon these developments, the field of computational humanities increasingly favors hybrid paradigms that synthesize deterministic domain knowledge with statistical machine learning. While pure machine learning models can struggle with strict mathematical constraints—such as rigid metrical syllable counts—rule-based engines lack adaptability to noise. Integrating transformer-based sequence scoring and edit-distance retrieval algorithms with deterministic classification pipelines enables context-aware candidate generation (Hellwig, 2020; Sandhan et al., 2021). In such a framework, the probabilistic model functions as an error corrector and nearest-match suggestion engine for degraded inputs, while the deterministic core maintains exact metrical verification for valid sequences. This hybrid approach significantly improves overall system resilience and expands the practical utility of computational prosody tools across real-world, digitized text corpora.

3. Methodology

3.1 Data Collection & Curation

The dataset for training and evaluating the proposed model is compiled from two complementary sources to balance volume and ground-truth reliability:

- **Primary Sources (High-Precision Ground Truth):** Selected classical verses will be extracted from authoritative digital repositories, specifically **Vedabase.io** (*Bhagavad-gītā* and *Śrīmad-Bhāgavatam*). These provide standardized Devanagari text and Roman transliteration (IAST), serving as clean benchmarks.
- **Digital Corpus of Sanskrit (DCS):** Large-scale, metrically annotated open-access text dumps are integrated from the DCS repository to ensure structural diversity across broad classical literature (*kāvya* and *Purāṇas*).

3.2 Core Engine Implementation

The original *Chandojñānam* rule-based parsing logic will be implemented in Python to establish a clean execution base and baseline performance benchmark.

3.3 ML-Driven Verse Corrector (Masked Language Model)

- **Architecture:** Fine-tuning a pre-trained transformer model (e.g., SanskritBERT) using a Masked Language Model objective.
- **Corruption Strategy:** Synthetic noise (substitutions, deletions, insertions of *Laghu/Guru* syllables) will be injected into training verses to train the model in reconstructing plausible meter-compliant sequences.
- **Nearest Match Engine:** Probabilistic candidate generation combined with sequence edit-distance algorithms to output ranked k-nearest *Chanda* recommendations for degraded input.

3.4 Evaluation Metrics

- **Exact Match Accuracy:** Evaluation on clean test sets.
- **Top-k Categorical Accuracy:** Measurement of correct *Chanda* presence within the top-1, top-3, and top-5 suggestions for noisy input.
- **Robustness Index:** Performance degradation rate across incremental artificial noise levels (10%, 20%, 30% corruption).

4. Expected Outcomes

- A fine-tuned Masked Language Model specifically adapted for verse reconstruction and error correction in metrical text.

- A comprehensive comparative benchmark proving the improved resilience of hybrid ML approaches over traditional rule-based parsers.

5. Work Plan of the Study

The proposed research is designed to be completed within a 3 month timeframe. The following Gantt chart outlines the distribution of tasks from initial data parameterization to final documentation.

- **Phase 1: Domain & Foundational Analysis (Weeks 1–3)**
    - Study classical *Chanda Śāstra* principles (*Laghu/Guru* rules, *Gaṇa* classifications).
    - Conduct deep model study (*Chandojñānam* baseline, Masked Language Models, SanskritBERT architecture).
- **Phase 2: Data Curation & Core Engine Implementation (Weeks 4–6)**
    - Data collection (DCS, Vedabase) and preprocessing
    - Implement a baseline rule-based engine.
    - Develop synthetic noise generation pipeline for model evaluation.
- **Phase 3: Model Fine-Tuning & Integration (Weeks 7–9)**
    - MLM adaptation: Fine-tune SanskritBERT on corrupt/masked metrical data.
    - Develop nearest-match module (edit-distance & probabilistic candidate scoring).
    - System integration & Web App development for visualization.
- **Phase 4: Evaluation & Documentation (Weeks 10–12)**
    - Perform comparative benchmarking against the baseline engine.
    - Thesis report writing and revisions.
    - Final thesis submission and defense presentation preparation.

Activities	Timeline(12 weeks / 3months)											
	1	2	3	4	5	6	7	8	9	10	11	12
Phase 1: Domain & Foundation Study												
Study classical Chanda Śāstra principles (Laghu/Guru rules, Gaṇa classifications)												
Model study (Chandojñānam baseline, Masked Language Models, SanskritBERT architecture)												
Phase 2: Data Curation & Core Engine Implementation												
Data collection & preprocessing												
Implement baseline rule-based engine												
Synthetic noise generation												
Phase 3: Model Fine-Tuning & Integration												
MLM adaptation: Fine-tune SanskritBERT												
Develop nearest-match module												
System integration & Web App development												
Phase 4: Evaluation & Documentation												
Compare benchmarking against baseline engine												
Thesis report writing and revisions												
Final submission												

Figure 1: Gantt Chart for Research Completion (3-Month Schedule)

References

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